AIM : To configure a cryptocurrency wallet using MetaMask, connect it to the Sepolia test network, and integrate the wallet with a decentralized application (DApp) to perform and verify a test ETH transaction.

THEORY :

A cryptocurrency wallet is a digital application used to manage blockchain accounts and interact with decentralized networks. It does not directly store physical coins; instead, it stores and manages the cryptographic keys required to access blockchain assets.

MetaMask is a browser-based cryptocurrency wallet that supports Ethereum and other Ethereum-compatible blockchain networks. It provides users with a public wallet address for receiving assets and securely manages private keys for authorizing transactions. The public address can be shared, but the Secret Recovery Phrase and private key must never be shared.

A blockchain network is a distributed ledger maintained by multiple nodes. Ethereum provides a platform for executing smart contracts and decentralized applications. For development and testing, networks such as Sepolia are used. Sepolia is an Ethereum test network where test ETH has no real monetary value and is used to pay transaction gas fees during testing.

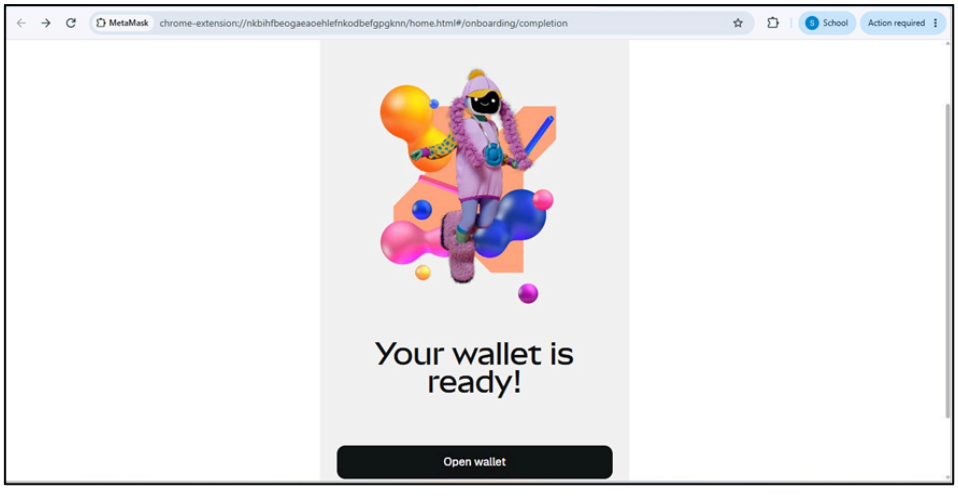
A DApp, or decentralized application, is an application that interacts with a blockchain through smart contracts. A DApp uses MetaMask to request permission to access the user's wallet address and to authorize blockchain transactions. The connection between the DApp and MetaMask is generally established through the Ethereum provider interface.

The **eth_accounts** method is used to retrieve the wallet address connected to the DApp. The DApp can also request permission to connect to the wallet. However, MetaMask does not expose the user's private key to the DApp. Instead, the DApp sends transaction requests to MetaMask, and the user reviews and approves them.

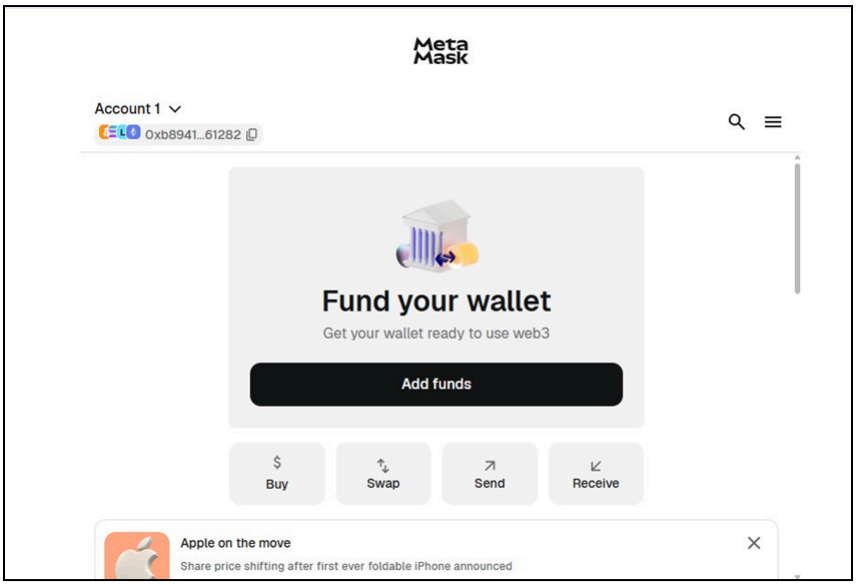
An EIP-1559 transaction is an Ethereum transaction format that includes a base fee and a priority fee. The base fee is determined by the network, while the priority fee acts as a tip to the validator. MetaMask displays the transaction details and estimated gas fee before the user confirms the transaction.

In this experiment, MetaMask was configured with the Sepolia test network, test ETH was obtained from a faucet, and the wallet was connected to the MetaMask Test DApp. An EIP-1559 transaction was then authorized through MetaMask and verified in the wallet's activity section.

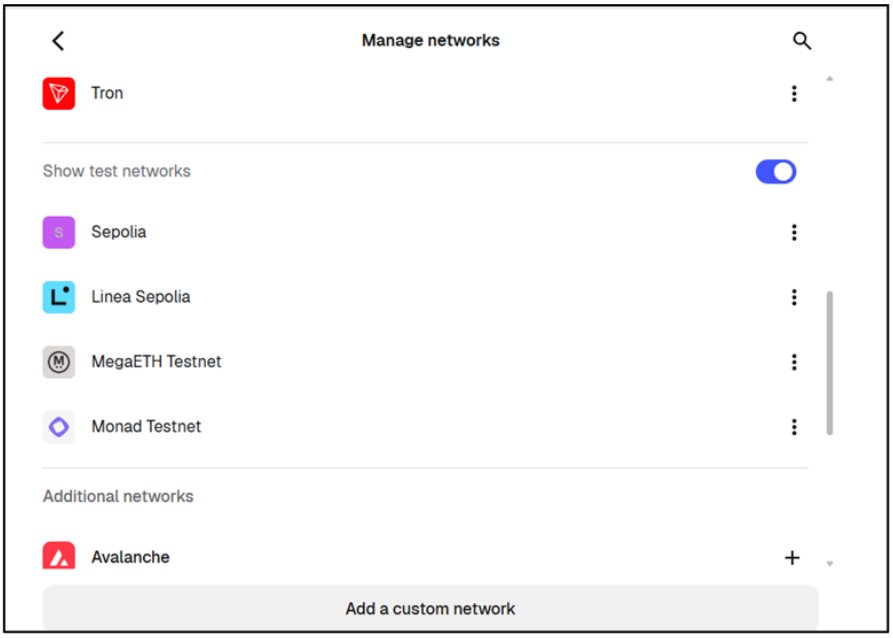
MetaMask Wallet Successfully Created and Ready to Use



MetaMask Wallet Dashboard Showing Account and Funding Options



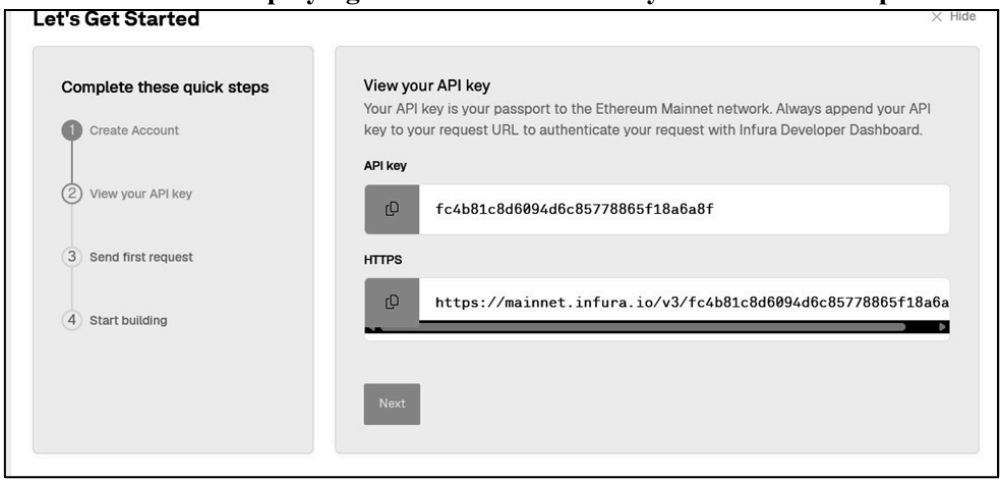
MetaMask Manage Networks Interface Showing Sepolia Test Network



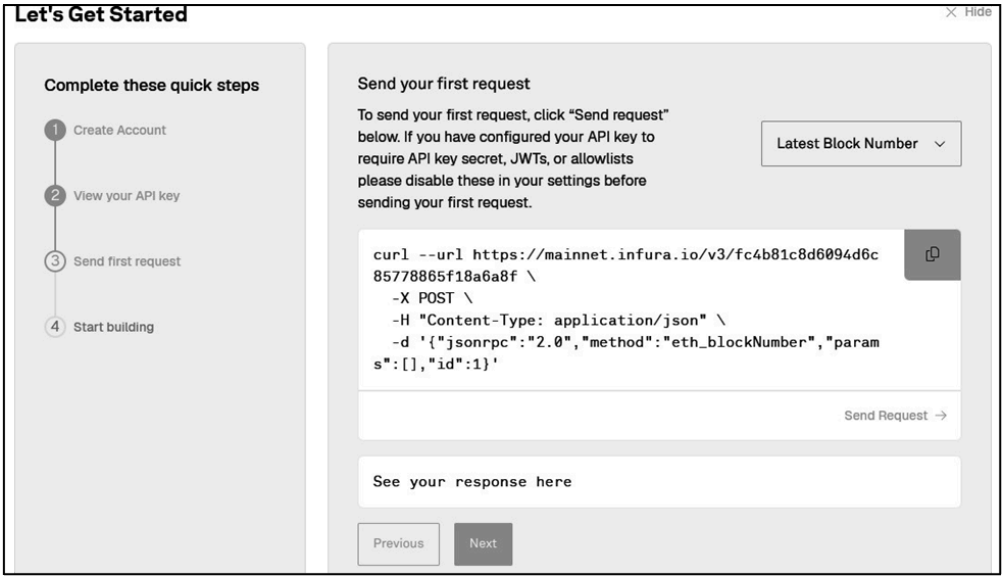
Infura Dashboard Showing Quick-Start Steps for API Key Configuration



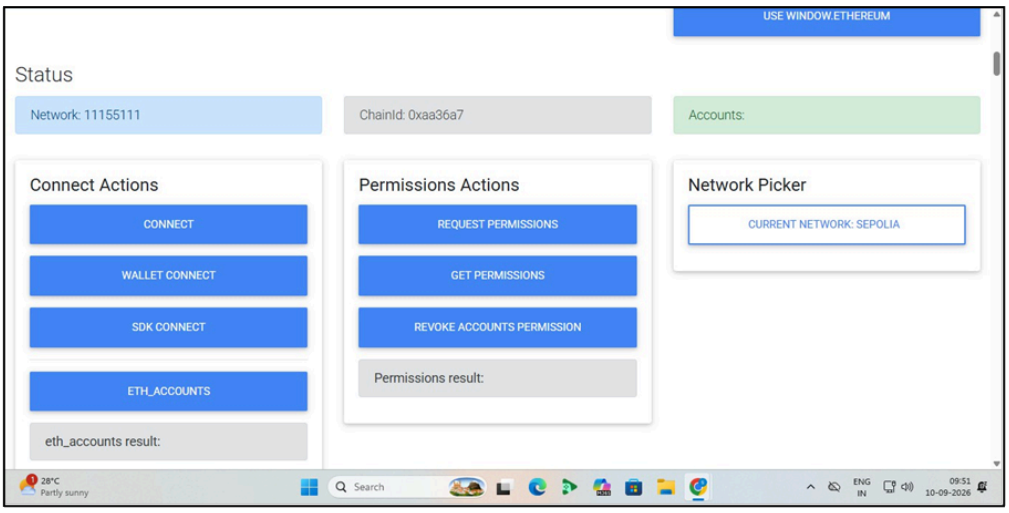
Infura Dashboard Displaying the Generated API Key and HTTPS Endpoint



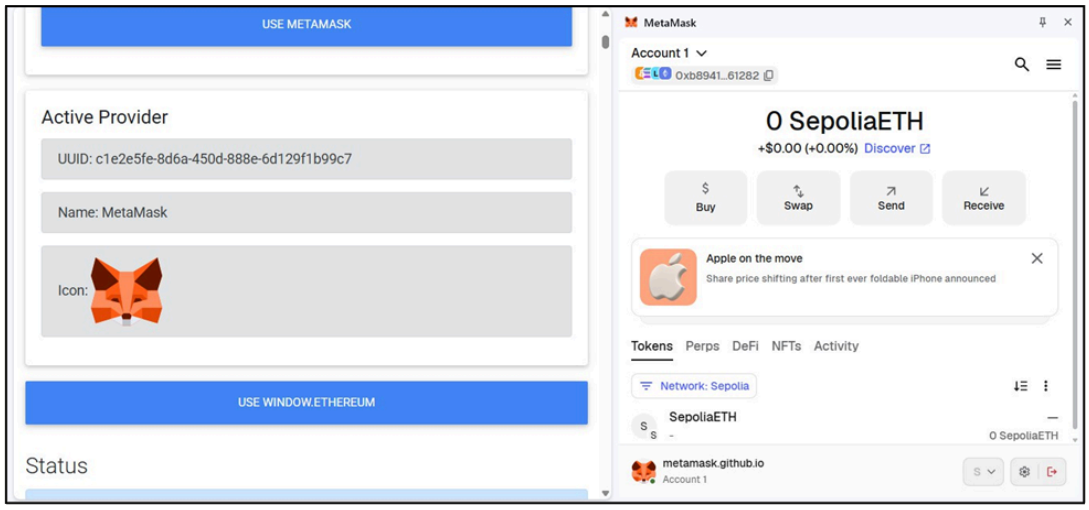
Infura Dashboard Showing the "Send Your First Request" Step



MetaMask Test DApp Showing Sepolia Network and Connection Actions



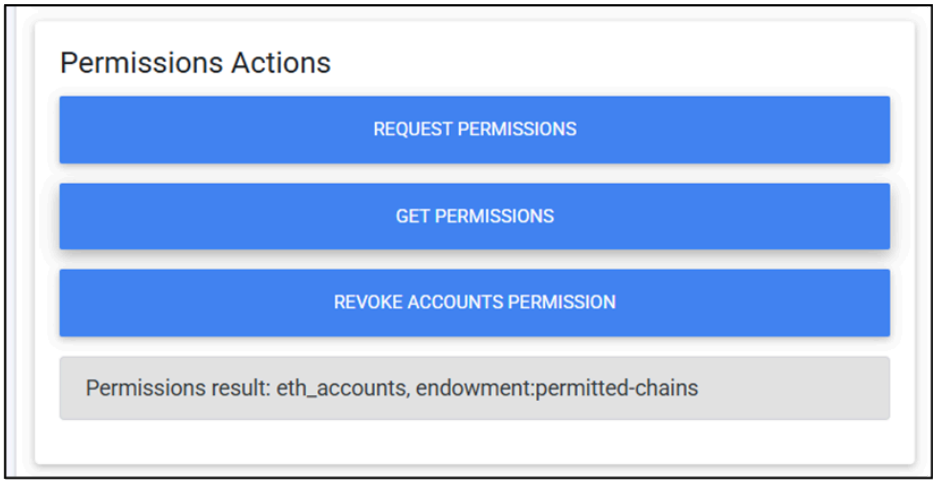
MetaMask Test DApp Showing Active MetaMask Provider and Sepolia Wallet Connection



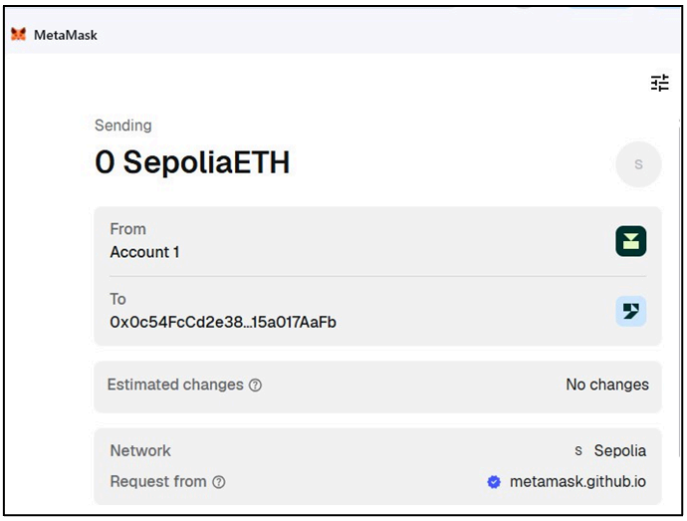
MetaMask Test DApp Successfully Connected and Wallet Address Retrieved Using eth_accounts



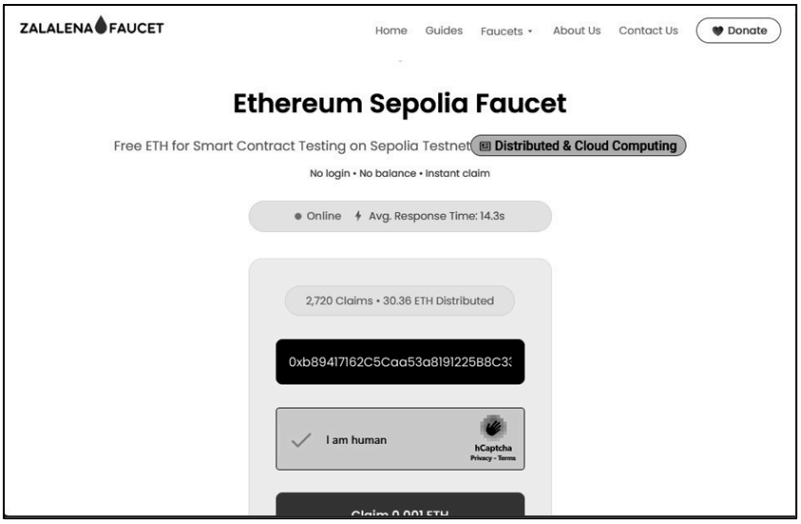
DApp Permissions Result Showing Authorized Account and Permitted Chains



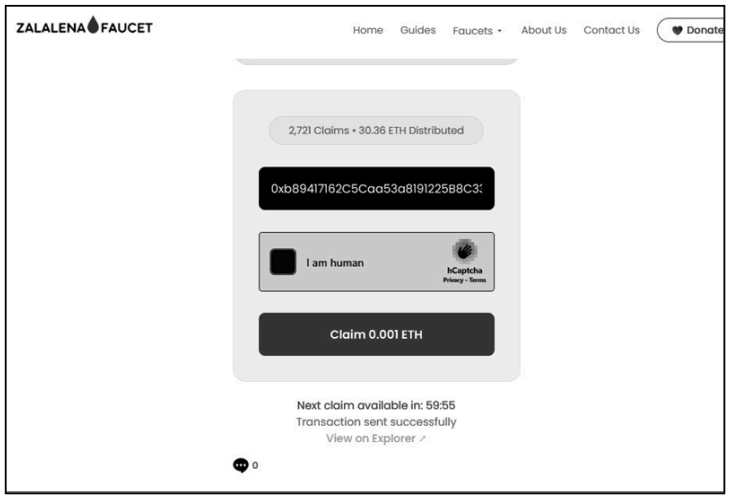
MetaMask Transaction Confirmation Request on the Sepolia Test Network



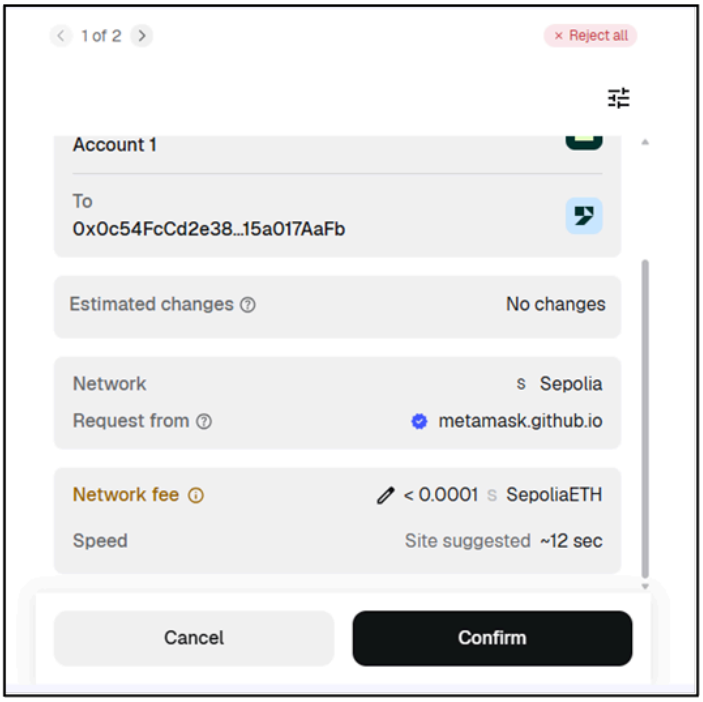
Requesting Sepolia Test ETH from a Cryptocurrency Faucet



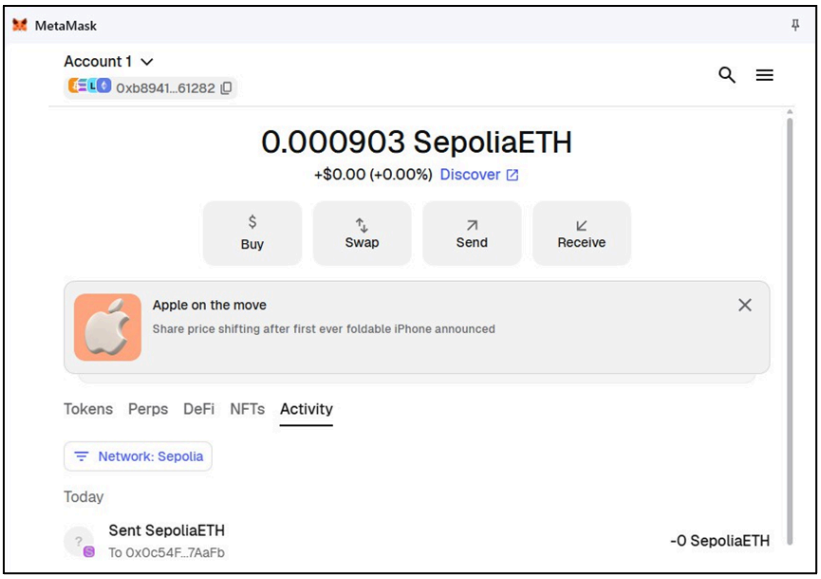
Successful Sepolia Test ETH Claim from Zalalena Faucet



MetaMask Transaction Confirmation Screen for an EIP-1559 Transaction on Sepolia



MetaMask Activity Tab Showing Successful Sepolia ETH Transaction



CONCLUSION :

The cryptocurrency wallet was successfully created and configured using MetaMask. The Sepolia test network was added, and test ETH was obtained through a faucet. The wallet was then connected to the MetaMask Test DApp, where the account address and permissions were successfully retrieved. Finally, an EIP-1559 transaction was authorized and displayed in the MetaMask Activity tab, demonstrating successful interaction between the wallet, DApp, and blockchain test network.